Def. Let (X,d) be a Metric space.
A sequence in $\{p_n\}$ is said to be Converge if: there exists a point $p \in X$ s.t
For all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow d(p_n, p) < \varepsilon$.

Def. Let $\{p_n\}$ be a Sequence in a Metric Space (X,d).
For some strictly Natural numbers sequence $\{n_i\}$,
$\{p_{n_i}\}$ is called subsequence of $\{p_n\}$.

Thm. Every Sequence in Compact Metric Space has a Convergent subsequence.
Proof. Let $\{p_n\}$ be a sequence in a Compact Metric space X.
If the range of $\{p_n\}$ is finite, then there is a $m \in \mathbb{N}$ s.t $Im\{p_n\} = \{p_1, p_2, \cdots, p_m\}$
Define for each $i \in \mathbb{N}$,

$$D_i = \{k \in \mathbb{N} \mid p_k = p_i\}$$

Then, for some $1 \leq j \leq m$, D_j is in infinite; because all that set is finite, then contradict to

$$\mathbb{N} = \bigcup_{i=1}^{m} D_i.$$

Now, we have a strictly increasing sequence $\{n_i\}$ in D_j, $\{p_{n_i}\}$ Converges to p_j.
If the range of $\{p_n\}$ is infinite, then the range has a limit point p. (By Lemma)
Now, for any $i \in \mathbb{N}$, we can choose n_i s.t $d(p_{n_i}, p) < \frac{1}{i}$
because $B_{\frac{1}{i}}(p)$ has infinitely many points of range of $\{p_n\}$,

Lemma. Infinite Subset of Compact set has a limit point.
proof. Let K be a Compact set, and $E \subseteq K$ be an infinite subset.
If any point q of K is not limit point of E, then there is an open $V_q \subseteq K$ s.t
$E \cap (V_q \setminus \{q\}) = \emptyset$.
Now, $\{V_q \mid q \in K\}$ is open Cover of K, thus there exists a finite subcover
$\{V_{q_i} \mid i = 1, 2, \cdots, n\}$, s.t $K = \bigcup_{i=1}^{n} V_{q_i}$
But, $E = E \cap K = E \cap \left(\bigcup_{i=1}^{n} V_{q_i}\right) = \bigcup_{i=1}^{n} (E \cap V_{q_i})$ Can have finite element, Contradiction.

Thm. The subsequential limits of a sequence $\{p_n\}$ in a metric space (X,d) form a closed set.

Proof. Let E be the set of all subsequential limits of $\{p_n\}$, let $q \in E'$ be a limit point of E.
If $E=\phi$, then there is nothing to prove. Choose $n_1 \in \mathbb{N}$ s.t $p_{n_1} \neq q$.
(If $\forall n \in \mathbb{N}$, $p_n = q$, then $E = \{q\}$, there is nothing to prove)
Suppose that $n_1, \cdots, n_{i-1}$ are chosen. Since $q \in E'$, there is $p \in E$ s.t

$$d(p,q) < \frac{1}{2^i} \cdot d(p_{n_1}, q)$$

And since $p \in E$, there is $n_i > n_{i-1}$ s.t

$$d(p, p_{n_i}) < \frac{1}{2^i} d(p_{n_1}, q).$$

now,

$$d(q, p_{n_i}) \leq d(p,q) + d(p, p_{n_i}) < \frac{1}{2^{i-1}} \cdot d(p_{n_1}, q).$$

The sequence $\{p_{n_i}\}$ is subsequence of $\{p_n\}$ s.t $p_{n_i} \to q$ as $i \to \infty$.
Thus, $q \in E$.

Def. Let E be a non-empty subset of a Metric Space X.
Define a diameter of E as:

$$\operatorname{diam} E \overset{\text{def}}{=} \sup\{ d(p,q) \mid p,q \in E \}.$$

Thm. $\operatorname{diam} E = \operatorname{diam} \overline{E}$, for any subset E of a Metric Space X.

Proof. $\operatorname{diam} E \le \operatorname{diam} \overline{E}$ is clear. To show reverse inequality, let $\varepsilon > 0$ be given.

For any $p, q \in \overline{E}$, there exist $p', q' \in E$ s.t.

$$d(p,p'),\ d(q,q') < \varepsilon.$$

Now,

$$d(p,q) \le d(p,p') + d(p',q') + d(q',q) < 2\varepsilon + \operatorname{diam} E$$

This means

$$\operatorname{diam} \overline{E} \le 2\varepsilon + \operatorname{diam} E.$$

Since $\varepsilon > 0$ was chosen arbitrarily, thus $\operatorname{diam} \overline{E} \le \operatorname{diam} E$.

Thm. Let (X,d) be a Metric Space, $A, B \subseteq X$ be subsets.

「Sup, inf, lim sup, lim inf.

「Def. Let $(X, \leq)$ be a totally ordered set, $A \subseteq X$ be a subset.
A $\alpha \in X$ is called Supremum of A if:

1). $\forall s \in A$, $s \leq \alpha$
2). If $\beta \in X$ satisfies $\forall s \in A$, $s \leq \beta$, then $\alpha \leq \beta$.

「Lemma. Let $A \subseteq \mathbb{R}$ be a subset. TFAE:

a) $s = \sup A$
b) s is an upper bound of A, and for any $\varepsilon > 0$, there exists a $r \in A$ s.t. $s - \varepsilon < r \leq s$

「Prop. Let $A, B \subseteq \mathbb{R}$ be non-empty subsets.

1). $\sup(A+B) = \sup A + \sup B$, $\sup(AB) = \sup A \cdot \sup B$ if $A, B \subseteq (0, \infty)$
2) $\sup A = -\inf(-A)$
3) $\sup(A \cup B) = \max(\sup A, \sup B)$, $\sup(A \cap B) = \min(\sup A, \sup B)$

Proof. Let $\varepsilon > 0$ be given. Then, there exist $a \in A$ and $b \in B$ such that

$$\sup A + \sup B - 2\varepsilon < a + b \leq \sup A + \sup B.$$

Since $a+b$ lies in $A+B$, and clearly $\sup A + \sup B$ is an upper bound of $A+B$, thus $\sup A + \sup B = \sup(A+B)$.

To show 2), again, let $\varepsilon > 0$ be given. First, $-\sup A$ is lower bound of $-A$, because

$$\forall a \in A,\ a \leq \sup A \iff \forall a \in A,\ -a \geq -\sup A.$$

And, there exists $a \in A$ s.t.

$$\sup A - \varepsilon < a \leq \sup A.$$

This equals to

$$-\sup A \leq -a < -\sup A + \varepsilon.$$

Thus $-\sup A = \inf(-A)$.

Def. Let $\{a_n\}$ be a sequence of Real Numbers.

Define limit supremum and limit infimum as:

$$\limsup_{n\to\infty} a_n \overset{def}{=} \lim_{n\to\infty} \sup_{k\geq n} a_k \overset{(*)}{=} \inf_{n\in\mathbb{N}} \sup_{k\geq n} a_k. \quad \left(\text{if } \{a_n\} \text{ is not bounded above, define } +\infty\right)$$

$$\liminf_{n\to\infty} a_n \overset{def}{=} \lim_{n\to\infty} \inf_{k\geq n} a_k \overset{(*)}{=} \sup_{n\in\mathbb{N}} \inf_{k\geq n} a_k.$$

Note: The useful fact $(*)$ is given by the following lemmas:

Lemma. $\{\sup_{k\geq n} a_k\}$ is monotonically-decreasing sequence.

Proof. It is clear, because $n_1 < n_2 \Rightarrow \{a_k \mid k \geq n_1\} \supseteq \{a_k \mid k \geq n_2\}$.

Lemma. If $\{b_n\}$ is monotonically increasing, then $\lim_{n\to\infty} b_n = \sup_{n\in\mathbb{N}} b_n$

Proof. If $\{b_n\}$ is not bounded above, there is nothing to prove

Assume $\{b_n\}$ is bounded above. Let $\varepsilon > 0$ be given. Then, there exists $N\in\mathbb{N}$ s.t

$$\sup_{n\in\mathbb{N}} b_n - \varepsilon < b_N \leq \sup_{n\in\mathbb{N}} b_n$$

Since $\{b_n\}$ is increasing sequence, for all $k\geq N$,

$$\sup_{n\in\mathbb{N}} b_n - \varepsilon < b_N \leq b_k. \quad \inf \quad \alpha \leq \sup_{k\geq n} a_k \leq \alpha + \varepsilon$$

Meanwhile, By definition, for all $k\in\mathbb{N}$,

$$b_k \overset{def}{\leq} \sup_{n\in\mathbb{N}} b_n < \sup_{n\in\mathbb{N}} b_n + \varepsilon \qquad -\varepsilon < \sup_{k\geq n} a_k - \alpha < \varepsilon$$

In sum,

$$k\geq N \Rightarrow \left|\sup b_n - b_k\right| < \varepsilon.$$

Note: limit supremum is independent of finite terms in the sequence.

Lemma. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$.

$\alpha = \limsup_{n\to\infty} a_n$ if and only if:

1) $\forall \varepsilon > 0, \exists N\in\mathbb{N}$ s.t $n \geq N \Rightarrow a_n < \alpha + \varepsilon$

2) $\forall \varepsilon > 0, \forall n\in\mathbb{N}, \exists k \geq n$ s.t $\alpha - \varepsilon < a_k$.

Thm. Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be real number sequences. Then,

1) If for all $n \in \mathbb{N}$, $a_n \le b_n$, then $\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} b_n$.

2) $\limsup_{n\to\infty}(a_n+b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$

3) If for all $n \in \mathbb{N}$, $a_n, b_n \ge 0$, then $\limsup_{n\to\infty}(a_n b_n) \le (\limsup_{n\to\infty} a_n)\cdot(\limsup_{n\to\infty} b_n)$

4). $\limsup_{n\to\infty} a_n = -\liminf_{n\to\infty}(-a_n)$

5). If for all $n \in \mathbb{N}$, $a_n > 0$, then $\limsup_{n\to\infty} a_n = \left(\liminf_{n\to\infty} a_n^{-1}\right)^{-1}$

6). If for all $n \in \mathbb{N}$, $c_n \ge 0$, then

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \le \liminf_{n\to\infty} c_n^{\frac{1}{n}} \le \limsup_{n\to\infty} c_n^{\frac{1}{n}} \le \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$$

Proof. 1). Let $\varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ s.t.

$$n \ge N \Rightarrow \limsup_{n\to\infty} a_n - \varepsilon < a_n$$

$a + r$

$b + r$

$$ab + r(a+b) + r^2 = ab + \varepsilon.$$

$$r(a+b) + r^2 = \varepsilon$$

$$\Leftrightarrow r = \frac{-(a+b) \pm \sqrt{(a+b)^2 + 4\varepsilon}}{2}.$$

$$\limsup a_n \cdot a_n^{-1} = \limsup 1 = 1 \le \limsup a_n \cdot \limsup a_n^{-1}$$

$$= (\limsup a_n^{-1})^{-1} \le \limsup a_n$$

Put $\alpha = \liminf_{n\to\infty} \frac{C_{n+1}}{C_n}$. If $\alpha = 0$, then there is nothing to prove.

Assume $\alpha > 0$. Let $\alpha > \varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow \frac{C_{n+1}}{C_n} \geq \alpha - \varepsilon$$

That is, for $n \geq N$,

$$C_n \geq (\alpha-\varepsilon)^{n-N} \cdot C_N \Rightarrow C_n^{\frac{1}{n}} \geq (\alpha-\varepsilon)^{1-\frac{N}{n}} \cdot C_N^{\frac{1}{n}} \to \alpha - \varepsilon.$$

This implies

$$\liminf_{n\to\infty} C_n^{\frac{1}{n}} \geq \alpha = \liminf_{n\to\infty} \frac{C_{n+1}}{C_n}.$$

Test.

Root Test. Given $\sum a_n$, let $\alpha = \limsup_{n\to\infty} \sqrt[n]{|a_n|}$.

1) If $\alpha < 1$, $\sum a_n$ Converges absoluteely.

2) If $\alpha > 1$, $\sum a_n$ diverges

proof. Assume $\alpha < 1$. Then, for β s.t $\alpha < \beta < 1$, there exists $N \in \mathbb{N}$ s.t

$$k \geq N \Rightarrow \sqrt[k]{|a_k|} < \alpha + (\beta - \alpha) = \beta \Rightarrow |a_k| < \beta^k$$

Now, for any $n \geq N$,

$$\sum_{k=N}^{n} |a_k| \leq \sum_{k=N}^{n} \beta^k = \beta^N \cdot \frac{1-\beta^{n-N+1}}{1-\beta} \rightarrow \frac{\beta^N}{1-\beta} \quad \text{as } n \to \infty$$

Therefore,

$$\sum_{k=N}^{\infty} |a_k| < \infty.$$

Assume $\alpha > 1$, Then, for β s.t $\alpha > \beta > 1$, there exist infinitely many $k \in \mathbb{N}$ s.t

$$\sqrt[n]{|a_n|} > \beta > 1 \Rightarrow |a_n| > 1.$$

Thus $\sum a_n$ diverges,

Ratio Test. Given $\sum a_n$, let $\alpha = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$.

1) If $\alpha < 1$, $\sum a_n$ Converges absoluteely

2) If $\alpha > 1$, $\sum a_n$ diverges.

proof. Assume $\alpha < 1$. Then, for β s.t $\alpha < \beta < 1$, there exists $N \in \mathbb{N}$ s.t

$$k \geq N \Rightarrow \left|\frac{a_{k+1}}{a_k}\right| < \beta \Rightarrow |a_{k+1}| < \beta |a_k|.$$

Now, for any $n \geq N$,

$$|a_n| < \beta |a_{n-1}| < \beta^2 |a_{n-2}| < \cdots < \beta^{n+N-1} |a_{n-(n+N-1)}| < \beta^{n+N} |a_N|$$

This gives the result.